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Coffee brewing device and method of use

Abstract

A coffee brewing device allowing for easy post-brewing cleanup includes a cup, which is positionable in a container to which a lid is attachable to close its top. Hot water and coffee grounds can be added to the container. A handle, which is rotationally attached to an upper end of a tube that is slidably positioned through a hole in the lid, is selectively attachable to the rod, which extends from the cup. The rod is insertable through a filter and into the tube to position the filter adjacent to the lid. A user pushes down on the handle to force the filter through the slurry so that the coffee grounds collect in the cup. The user then attach the handle to the rod to remove the lid, the filter, the cup, and the coffee grounds from the container.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(2) Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

(3) Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

(4) Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

(5) Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

(6) The disclosure relates to brewing devices and more particularly pertains to a new brewing device allowing for easy post-brewing cleanup. The present invention discloses a brewing device having a cup for collection of coffee grounds, thereby allowing the cup and coffee grounds to be removed from the container for facilitating cleanup.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

(7) The prior art relates to brewing devices, which may comprise containers having detachable bottom sections that collect coffee grounds, cups positionable at bottoms of containers, wherein an upper limit of the cup may engage a filter module upon depression of the filter modules, What is lacking in the prior art is a brewing device having a cup positionable at a bottom of a container that does not couple to a filter module. Rather, the filter module can be pushed down to the upper limit of the cup, or even into the cup, prior to the cup, coffee grounds, and filter module being removed from the container by means of a rod extending through the filter module and a tube to which the filter module is attached.

BRIEF SUMMARY OF THE INVENTION

(8) An embodiment of the disclosure meets the needs presented above by generally comprising a cup, which is circumferentially complementary to and selectively positionable in a container. A rod, which has a lower terminus attached to the cup, extends axially from the cup and protrudes through a top of the container with the cup positioned on a bottom of the container. The container is configured for addition of hot water and coffee grounds. A lid, through which a hole is axially positioned, is selectively attachable to the container to substantially close the top of the container. A handle, which is rotationally attached to an upper end of a tube that is slidably positioned through the hole, is selectively attachable to an upper terminus of the rod. A filter module is attached to a lower end of the tube. The rod is insertable through the filter module and into the tube to position the filter module adjacent to the lid and atop a slurry of the hot water and coffee grounds. A user is positioned to push down on the handle to force the filter module through the slurry so that the coffee grounds collect in the cup. The user then is positioned to attach the handle to the rod and to remove the lid, the filter module, the cup, and the coffee grounds from the container.

(9) Another embodiment of the disclosure includes a method of brewing coffee. The method comprises provision steps, which entail providing a coffee brewing device, according to the disclosure above and providing coffee grounds and hot water. Operational steps of the method are adding the coffee grounds to the cup, positioning the cup on the bottom of the container, adding the hot water to the container, positioning tube over the rod, pushing down on the handle for forcing the filter module through the slurry, attaching the handle to the rod, dispensing the coffee from the container, and extracting the lid, the filter module, the cup, and the coffee grounds from the container.

(10) There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

(11) The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

(1) The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

(2) FIG. 1 is an isometric perspective view of a coffee brewing device according to an embodiment of the disclosure.

(3) FIG. 2 is an exploded view of an embodiment of the disclosure.

- (4) FIG. 3 is a front view of an embodiment of the disclosure.
- (5) FIG. 4 is an exploded view of an embodiment of the disclosure.
- (6) FIG. 5 is a cross-sectional view of an embodiment of the disclosure.
- (7) FIG. 6 is a flow diagram for a method utilizing an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(8) With reference now to the drawings, and in particular to FIGS. 1 through 6 thereof, a new brewing device embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

(9) As best illustrated in FIGS. 1 through 6, the coffee brewing device **10** generally comprises a cup **12**, which is circumferentially complementary to and selectively positionable in a container **14**. The container **14** is configured for addition of hot water and coffee grounds. Many configurations for such brewing devices are taught in the prior art and are anticipated by the present invention. Such brewing devices typically are cylindrical, have handles, which are attached or attachable to their containers **14** and which allow for lifting and pouring of liquid contents, and may comprise glass, double-walled stainless steel, or the like.

(10) A rod **16**, which has a lower terminus **18** that is attached to the cup **12**, extends axially from the cup **12** and protrudes through a top **20** of the container **14** with the cup **12** positioned on a bottom **22** of the container **14**. The rod **16** is threaded adjacent to its upper terminus **24** and its lower terminus **18**. A coupler **98** is attached to the lower terminus **18** of the rod **16** and is configured to selectively couple to the cup **12** so that the rod **16** is removably attachable to the cup **12**.

(11) The coupler **98** may comprise a magnet **100**, or other coupling means, such as, but not limited to, threaded sockets, hooks, or the like. The magnet **100** can magnetically attach to the cup **12**, should the cup **12** comprise paramagnetic metal, such as stainless steel. As shown in FIGS. 2 and 5, a recess **102** is axially positioned in the magnet **100**. The recess **102** is internally threaded so that the lower terminus **18** of the rod **16** is selectively threadedly insertable into the recess **102** to removably attach the rod **16** to the magnet **100**. The coffee brewing device **10** also may comprise a disc **104**, which is attached to and centrally positioned in the cup **12**. The disc **104** is magnetized so that the disc **104** is positioned for magnetic attachment to the magnet **100**, thereby providing stronger coupling between the rod **16** and the cup **12**.

(12) A lid **106**, through which a hole **26** is axially positioned, is selectively attachable to the container **14** to substantially close the top **20** of the container **14**. A tube **28**, which is externally threaded adjacent to its lower end **30**, is slidably positioned through the hole **26**. A stop ring **32** is attached to and extends circumferentially around the tube **28** proximate to the lower end **30**.

(13) A handle **34** is rotationally attached to an upper end **36** of the tube **28**. The handle **34** also is selectively attachable to the upper terminus **24** of the rod **16**. The handle **34** may be tubular and internally threaded so that the upper terminus **24** of the rod **16** is selectively threadedly insertable into the handle **34** to removably attach the handle **34** to the rod **16**. Other reversible attachment means for attaching the handle **34** to the rod **16** also are anticipated, such as, but not limited to, friction coupling, spring-loaded pins, or the like.

(14) As shown in FIGS. 2-5, the coffee brewing device **10** also comprises a frame **38**, which is substantially rectangular. A first circular cutout **40** and a second circular cutout **42** are positioned in a lower member **44** and an upper member **46** of the frame **38**, respectively. A pair of first indents **48** extends into and circumferentially around the tube **28** proximate to the upper end **36**. The tube **28** is insertable into the first circular cutout **40** so that each first clip **50** of a pair of first clips **50** is selectively insertable into a respective first indent **48** to removably attach the tube **28** to the frame **38**. A second indent **52** extends into and circumferentially around the handle **34** proximate to a lower limit **54** of the handle **34**. The handle **34** is selectively slidably positionable in the second circular cutout **42** so that a second clip **108** is selectively insertable into the second indent **52** to slidably attach the handle **34** to the frame **38**.

(15) A filter module **56** is attached to the lower end **30** of the tube **28**. The rod **16** is insertable through the filter module **56** and into the tube **28** to position the filter module **56** adjacent to the lid **106** and atop a slurry of the hot water and coffee grounds in the container **12**. A user is positioned to push down on the handle **34** to force the filter module **56** through the slurry so that the coffee grounds collect in the cup **12**. The user then is positioned to attach the handle **34** to the rod **16** and to remove the lid **106**, the filter module **56**, the cup **12**, and the coffee grounds from the container **14**, which significantly simplifies post-brewing cleanup relative to prior art brewing devices. The coffee brewing device **10** can be used to make coffee, tea, hot chocolate, cold brews, fruit infusions, herbal drinks, and the like.

(16) The prior art teaches many configurations of filter modules **56** for such brewing devices and these are anticipated by the present invention. One example of a filter module **56** comprises an upper disc **58**, a filter disc **60**, and a lower disc **62**, as shown in FIGS. **2** and **4**. A plurality of first orifices **64**, a second orifice **66**, and a plurality of third orifices **68** are positioned in the upper disc **58**, the filter disc **60**, and the lower disc **62**, respectively. A central one **70** of the plurality of first orifices **64** is complementary to the tube **28** so that the lower end **30** of the tube **28** is insertable into the central one **70** of the first orifices **64** to abut the upper disc **58** to the stop ring **32**. The second orifice **66** is centrally positioned in the filter disc **60** so that the lower end **30** of the tube **28** is insertable into the second orifice **66** to abut the filter to the upper disc **58**. A coupling ring **72**, which is internally threaded, is attached to the lower disc **62** so that the lower end **30** of the tube **28** is threadedly insertable into the coupling ring **72** to removably attach the filter module **56** to the tube **28**.

(17) In use, the coffee brewing device **10** enables a method of brewing coffee **74**, which comprises a first provision step **76** that entails providing a coffee brewing device **10**, according to the specification above. A second provision step **78** of the method **74** is providing coffee grounds and hot water. A first operational step **80** of the method **74** is adding the coffee grounds to the cup **12**. A second operational step **82** of the method **74** is positioning the cup **12** on the bottom **22** of the container **14**. A third operational step **84** of the method **74** is adding the hot water to the container **14** to generate a slurry of the hot water and coffee grounds. A fourth operational step **86** of the method **74** is positioning tube **28** over the rod **16**, such the filter module **56** is positioned atop the slurry and such that the lid **106** substantially closes the top **20** of the container **14**.

(18) A fifth operational step **88** of the method **74** is allowing the slurry to rest. A sixth operational step **90** of the method **74** is pushing down on the handle **34** to force the filter module **56** through the slurry. A seventh operational step **92** of the method **74** is attaching the handle **34** to the rod **16**. An eighth operational step **94** of the method **74** is dispensing the coffee from the container **14**. A ninth operational step **96** of the method **74** is extracting the lid **106**, the filter module **56**, the cup **12**, and the coffee grounds from the container **14**.

(19) With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

(20) Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility

that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A coffee brewing device comprising: a container; a cup circumferentially complementary to the container, such that the cup is selectively positionable in the container to rest upon a bottom of the container, wherein the container is configured for addition of hot water and coffee grounds; a rod having a lower terminus attached to the cup, such that the rod extends axially from the cup and protrudes through a top of the container with the cup positioned on the bottom of the container; a lid selectively attachable to the container for substantially closing the top of the container; a hole axially positioned through the lid; a tube slidably positioned through the hole; a handle rotationally attached to an upper end of the tube, the handle being selectively attachable to an upper terminus of the rod; and a filter module attached to a lower end of the tube, such that the rod is insertable through the filter module into the tube for positioning the filter module adjacent to the lid and atop a slurry of the hot water and coffee grounds, positioning a user for pushing down on the handle for forcing the filter module through the slurry, such that the coffee grounds collect in the cup, positioning the user for attaching the handle to the rod and for removing the lid, the filter module, the cup, and the coffee grounds from the container.
2. The coffee brewing device of claim 1, further including a coupler attached to the lower terminus of the rod and being configured for selectively coupling to the cup, such that the rod is removably attachable to the cup.
3. The coffee brewing device of claim 2, wherein the coupler comprises a magnet.
4. The coffee brewing device of claim 3, further including: the rod being threaded adjacent to the lower terminus of the rod; and a recess axially positioned in the magnet, the recess being internally threaded, such that the lower terminus of the rod is selectively threadedly insertable into the recess for removably attaching the rod to the magnet.
5. The coffee brewing device of claim 3, further including a disc attached to and centrally positioned in the cup, the disc being magnetized, such that the disc is positioned for magnetic attachment to the magnet.
6. The coffee brewing device of claim 1, wherein: the rod is threaded adjacent to the upper terminus of the rod; and the handle is tubular and internally threaded, such that the upper terminus of the rod is selectively threadedly insertable into the handle for removably attaching the handle to the rod.
7. The coffee brewing device of claim 1, further including: a frame, the frame being substantially rectangular; a first circular cutout positioned in a lower member of the frame; a pair of first indents extending into and circumferentially around the tube proximate to the upper end, such that the tube is insertable into the first circular cutout; a pair of first clips, each first clip of the pair of first clips being selectively insertable into a respective first indent of the pair of first indents for removably attaching the tube to the frame; a second circular cutout positioned in an upper member of the frame, such that the handle is selectively slidably positionable in the second circular cutout; a second indent extending into and circumferentially around the handle proximate to a lower limit of the handle; and a second clip selectively insertable into the second indent for slidably attaching the handle to the frame.
8. The coffee brewing device of claim 1, further including: the tube being externally threaded adjacent to a lower end of the tube; a stop ring attached to and extending circumferentially around the tube proximate to the lower end; and the filter module comprising: an upper disc, a plurality of first orifices positioned in the upper disc, a central one of the plurality of first orifices being complementary to the tube, such that the lower end of the tube is insertable into the central one of the plurality of first orifices to abut the upper disc to the stop ring, a filter disc, a second orifice

centrally positioned in the filter disc, such that the lower end of the tube is insertable into the second orifice to abut the filter disc to the upper disc, a lower disc, a plurality of third orifices positioned in the lower disc, and a coupling ring attached to the lower disc, the coupling ring being internally threaded, such that the lower end of the tube is threadedly insertable into the coupling ring for removably attaching the filter module to the tube.

9. A coffee brewing device comprising: a container; a cup circumferentially complementary to the container, such that the cup is selectively positionable in the container to rest upon a bottom of the container, wherein the container is configured for addition of hot water and coffee grounds; a rod having a lower terminus attached to the cup, such that the rod extends axially from the cup and protrudes through a top of the container with the cup positioned on the bottom of the container, the rod being threaded adjacent to an upper terminus and the lower terminus of the rod; a coupler attached to the lower terminus of the rod and being configured for selectively coupling to the cup, such that the rod is removably attachable to the cup, the coupler comprising a magnet; a recess axially positioned in the magnet, the recess being internally threaded, such that the lower terminus of the rod is selectively threadedly insertable into the recess for removably attaching the rod to the magnet; a disc attached to and centrally positioned in the cup, the disc being magnetized, such that the disc is positioned for magnetic attachment to the magnet; a lid selectively attachable to the container for substantially closing the top of the container; a hole axially positioned through the lid; a tube slidably positioned through the hole, the tube being externally threaded adjacent to a lower end of the tube; a stop ring attached to and extending circumferentially around the tube proximate to the lower end; a handle rotationally attached to an upper end of the tube, the handle being selectively attachable to the upper terminus of the rod, the handle being tubular and internally threaded, such that the upper terminus of the rod is selectively threadedly insertable into the handle for removably attaching the handle to the rod; a frame, the frame being substantially rectangular; a first circular cutout positioned in a lower member of the frame; a pair of first indents extending into and circumferentially around the tube proximate to the upper end, such that the tube is insertable into the first circular cutout; a pair of first clips, each first clip of the pair of first clips being selectively insertable into a respective first indent of the pair of first indents for removably attaching the tube to the frame; a second circular cutout positioned in an upper member of the frame, such that the handle is selectively slidably positionable in the second circular cutout; a second indent extending into and circumferentially around the handle proximate to a lower limit of the handle; a second clip selectively insertable into the second indent for slidably attaching the handle to the frame; and a filter module attached to a lower end of the tube, such that the rod is insertable through the filter module into the tube for positioning the filter module adjacent to the lid and atop a slurry of the hot water and coffee grounds, positioning a user for pushing down on the handle for forcing the filter module through the slurry, such that the coffee grounds collect in the cup, positioning the user for attaching the handle to the rod and for removing the lid, the filter module, the cup, and the coffee grounds from the container, the filter module comprising: an upper disc, a plurality of first orifices positioned in the upper disc, a central one of the plurality of first orifices being complementary to the tube, such that the lower end of the tube is insertable into the central one of the plurality of first orifices to abut the upper disc to the stop ring, a filter disc, a second orifice centrally positioned in the filter disc, such that the lower end of the tube is insertable into the second orifice to abut the filter disc to the upper disc, a lower disc, a plurality of third orifices positioned in the lower disc, and a coupling ring attached to the lower disc, the coupling ring being internally threaded, such that the lower end of the tube is threadedly insertable into the coupling ring for removably attaching the filter module to the tube.

10. A method of brewing coffee comprising the steps of: providing a coffee brewing device comprising: a container, a cup circumferentially complementary to the container, such that the cup is selectively positionable in the container to rest upon a bottom of the container, a rod having a lower terminus attached to the cup, such that the rod extends axially from the cup and protrudes

through a top of the container with the cup positioned on the bottom of the container, a lid selectively attachable to the container for substantially closing the top of the container, a hole axially positioned through the lid, a tube slidably positioned through the hole, a handle rotationally attached to an upper end of the tube, the handle being selectively attachable to an upper terminus of the rod, and a filter module attached to a lower end of the tube, such that the rod is insertable through the filter module into the tube for positioning the filter module adjacent to the lid atop the container; providing coffee grounds and hot water; adding the coffee grounds to the cup; positioning the cup on the bottom of the container; adding the hot water to the container to generate a slurry of the hot water and coffee grounds; positioning tube over the rod, such that the filter module is positioned atop the slurry and such that the lid substantially closes the top of the container; allowing the slurry to rest; pushing down on the handle for forcing the filter module through the slurry; attaching the handle to the rod; dispensing the coffee from the container; and extracting the lid, the filter module, the cup, and the coffee grounds from the container.
